FACULTY OF INFORMATION SCIENCE

BACHELOR OF INFORMATION SCIENCE (HONS) INFORMATION SYSTEM MANAGEMENT (CDIM262)

ADVANCED DEVELOPMENT WEBSITE MANAGEMENT SYSTEM (IMS566)

GROUP ASSIGNMENT:

E-KAP (ELEKTRONIK KELULUSAN AKTIVITI PELAJAR)

PREPARED BY:

NAME	STUDENT ID
NURUL SHARIEZA BINTI MUHAMMAD ZAHIR	2024273062
NURHIDAYAH MUHAMMAD ADZIM @ AJIUN	2024696368
NURUL AINA' DAMIA BINTI ZUBIR	2024272254
WAN NORSYAFIQAH BINI SHAHRIR	2024221294

GROUP: CDIM2624ST2

PREPARED FOR:

DR. MUHAMMAD ASYRAF BIN WAHI ANUAR

SUBMISSION DATE:

17th JULY 2026

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1.0 INTRODUCTION

1.1 Background of The Project

Figure 1: E-KAP (Kelulusan Aktiviti Pelajar) Logo.

E-KAP (Elektronik Kelulusan Aktiviti Pelajar) is a web-based system developed to simplify and improve the management of student activity approval processes in higher education institutions. Student activities such as academic programs, workshops, seminars, competitions and club events require official approval before they can be organized. In many institutions, these approval processes are still handled manually through paper forms, emails or separate digital documents. Such methods are often time consuming, prone to human error and difficult to monitor, resulting in delays, misplaced documents and inefficient communication between students and administrative staff.

To overcome these challenges, e-KAP serves as a centralized website that streamlines the entire application and approval process. Through the website, students can create an account, log in securely, complete the activity application form, upload the required supporting documents and monitor the status of their application online. At the same time, staff members can access the administrative portal to review submitted applications, verify supporting documents and approve or reject applications efficiently. By digitalizing the workflow, the system reduces paperwork, improves record management and ensures that all application data is stored securely in a structured database.

The e-KAP website is developed using PHP as the server side scripting language and MySQL as the database management system. Bootstrap is used to create a responsive and user-friendly interface, while Laragon and Xampp provide the local server environment for development and testing. Visual Studio Code is used as the primary code editor throughout the development process. The integration of these technologies enables the website to deliver a reliable, organized and accessible platform for managing student activity approval processes.

E-KAP is designed to improve the efficiency, accuracy and transparency of managing student activity approvals. By replacing manual procedures with an online system, the website enables students and staff to complete the approval process more effectively while reducing administrative workload. The development of this project also provides practical experience in website development, database design, user interface implementation and system integration, demonstrating the application of technical knowledge to solve real administrative challenges in an educational environment.

1.2 Problem Statement

The current student activity approval process in many higher education institutions is still conducted using manual procedures, which often require students to print application forms, complete them by hand and submit the documents physically to the relevant office for approval. This process is time-consuming and inconvenient, especially when multiple signatures or supporting documents are required. Students may need to make several visits to the office, while administrative staff spend considerable time managing paper documents, verifying information and organizing application records. Furthermore, manual documentation increases the risk of misplaced forms, incomplete submissions, data duplication and delays in the approval process. Students also have limited access to updates on the status of their applications, making communication between students and staff less efficient. There is a need for a website system that can digitize the application and approval process, and improve a more organized and transparent method of managing student activity approvals.

1.3 Objective

1. To develop a website e-KAP system that enables students to submit student activity approval applications electronically without using manual paper forms.
2. To improve the efficiency of the approval process by providing staff with a centralized platform to review, verify, approve or reject student activity applications.
3. To provide students with an online platform where they can upload supporting documents and monitor the status of their applications in a more organized and transparent manner.

1.4 Target User

The e-KAP (Kelulusan Aktiviti Pelajar) system is designed for two primary groups of users:

Students

Students are the main users of the system. They can register an account, log in securely, submit student activity approval applications, upload the required supporting documents and monitor the status of their applications online. The system provides students with a more convenient and efficient alternative to the traditional paper based application process.

Staff (Administrators)

Staff members are responsible for managing and reviewing student activity applications submitted through the system. They can access the administrative dashboard to verify application details, review uploaded documents, approve or reject applications and update the application status. The system helps staff organize application records more effectively while reducing paperwork and improving the overall efficiency of the approval process.

2.0 GITHUB REPOSITORY LINK

This is the project repository information:

Project Name: e-KAP (Kelulusan Aktiviti Pelajar)

Electronic student application management system.

GitHub Repository: https://github.com/2024696368-design/ekap.git

3.0 ENTITY RELATIONSHIP DIAGRAM (ERD)

The e-KAP database is organised around five main entities: Users, Clubs, Applications, Documents and Reviews. The Application table is the central entity because it connects each programme application to the student who created it and the club selected by the student. Each application may also have several document records and several review records. The administrator who records a decision is identified through the admin_id foreign key in the Review table.

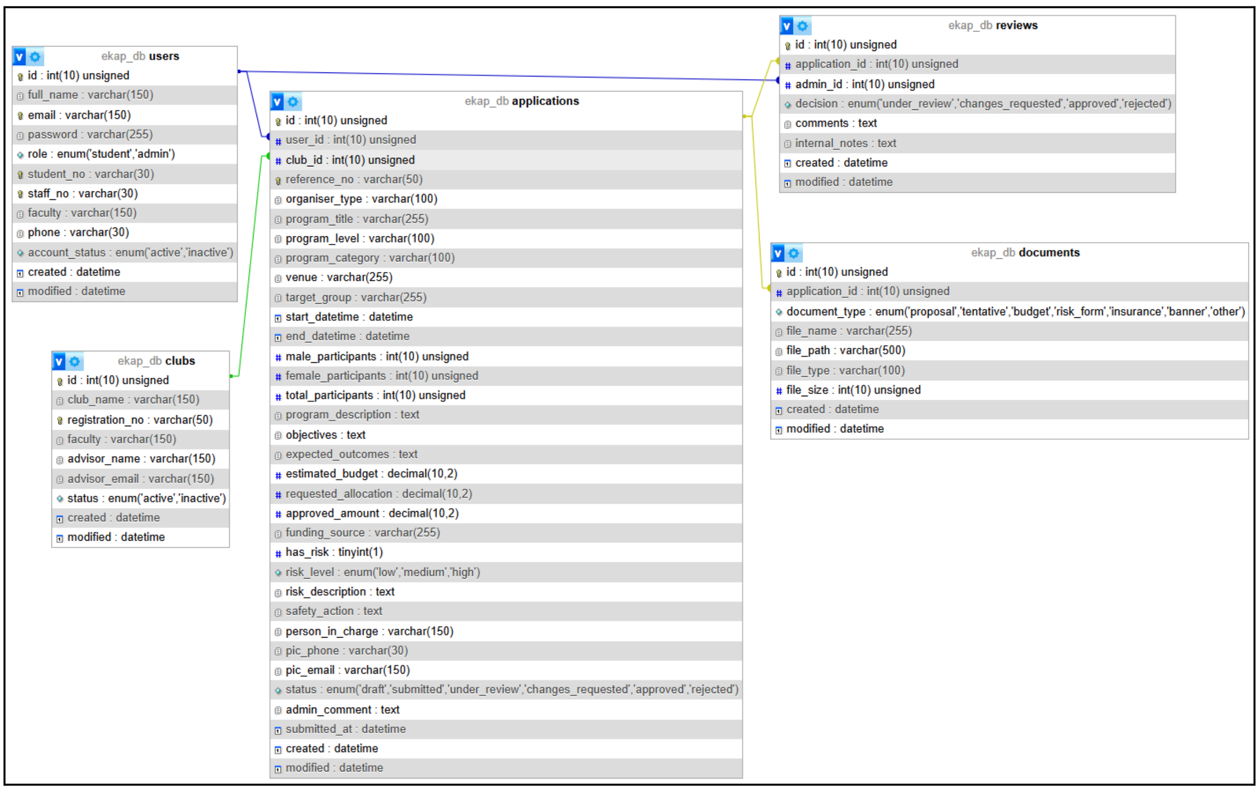

Figure 2: Entity Relationship Diagram for the e-KAP System

3.1 Entity Description

Entity	Purpose	Important Attributes
Users	Stores student and administrator account information	Id, full_name, email, password, role, student_no, staff_no, faculty, phone, account_status
Clubs	Stores registered club or society information that can be selected in programme application	id, club_name, registration_no, faculty, advisor_name, advisor_email, status
Applications	Stores the complete programme proposal, budget, risk, participant and contact information	id, user_id, club_id, reference_no, program_title, dates, venue, budget, risk details, status, approved_amount
Documents	Stores metadata for documents related to an application	id, application_id, document_type, file_name, file_path, file_type, file_size
Reviews	Stores each HEP review action and the administrator responsible for the description	id, application_id, admin_id, decision, comments, internal_notes

Table 1: Entity Description

3.2 Database Relationships

Relationship	Cardinality	Description
Users to Applications	One-to-many	One student account can create many programme applications. Each application belongs to one user.
Clubs to Applications	One-to-many	One registered club can be selected in many applications. Each application refers to one club.
Applications to Documents	One-to-many	One application can have several related document records. Each document belongs to one application
Applications to Reviews	One-to-many	One application may receive several review records as it progresses through the approval process
Users to Reviews	One-to-many	One administrator can record many review decisions. Each review is recorded by one administrator account

Table 2: Database Relationships

3.3 Key Business Rules

- A user email address must be unique.
- Student IDs and staff IDs must be unique when provided.
- A club registration number must be unique.
- An application must be connected to an existing user and an existing club.
- Students may only view and manage their own applications.
- Only draft and changes-requested applications may be edited by a student.
- Only draft applications may be deleted by a student.
- Only HEP administrators may review submitted applications.
- Comments are compulsory when an application is rejected or returned for changes.
- The approved allocation cannot exceed the allocation requested by the student.
- An approval letter is available only when the application status is approved.

4.0 SYSTEM REQUIREMENT

4.1 Software Requirement

Category	Tools/Technologies
Programming Language	PHP, HTML, CSS, JavaScript, Bootstrap 5
Browser	A current version of Google Chrome, Microsoft Edge or Mozilla Firefox
Database	MySQL, phpMyAdmin, CakePHP
Code Editor	Visual Studio Code
Framework	CakePHP 5.3.x
Local Server	Laragon, Xampp

Table 3: Software Requirements

4.2 Application Dependencies

Dependency	Version	Purpose
cakephp	5.3.*	Core MVC web application framework
cakephp/authentication	^4.2	Authentication - related package included in the project
cakephp/migrations	^5.0	Database migration support
dompdf	^3.1	Generates the official approval letter in PDF format
Bootstrap	5.3.3 (CDN)	Responsive layout and interface components
Bootstrap Icons	1.11.3 (CDN)	Interface icons
Three.js	r134 (CDN)	Required for the animated authentication background
Vanta.js NET	CDN	Animated visual background on login and registration pages
Google Font	CDN	Primary interface typeface

Table 4: Application Dependencies

4.3 Recommended Hardware Requirements

Item	Minimum	Recommended
Processor	Dual-core processor	Quad-core processor or better
Memory	4 GB RAM	8 GB RAM or more
Storage	1 GB free space for the application and dependencies	5 GB or more including backups and uploaded files
Network	Local network connection for development	Stable internet connection when CDN-based interface resources are used
Display	1366 x 768 resolution	1920 x 1080 resolution

Table 5: Hardware Recommendation

5.0 USER INTERFACE OVERVIEW

5.1 Overall Layout

The e-KAP interface uses a consistent layout across the system. A sticky navigation bar is displayed at the top of each standard page, while the footer identifies e-KAP and UiTM Puncak Perdana. The main interface uses Bootstrap 5, Bootstrap Icons, the Inter font and a custom green theme that matches the e-KAP identity. The layout is responsive, allowing forms, tables and navigation items to adjust for desktop, tablet and mobile screen sizes.

The system logo functions as a home link. After login, the user can access Dashboard, Applications and Clubs from the navigation bar. Student users also receive a New Application option, while administrators receive a User option. The profile area displays the user's name and role and provides access to the user profile and logout function.

5.2 Role-Based Navigation

Navigation Item	Student Access	Administrator Access	Purpose
Dashboard	Yes	Yes	Displays role-specific statistics, recent applications and quick-access links.
Applications	Own applications only	All applications	Lists and filters programme applications.
Clubs	View	Manage	Displays registered clubs; administrators can add, edit and delete eligible clubs.
New Application	Yes	No	Opens the programme application form.
Users	No	Yes	Allows administrators to manage and activate user accounts.
Profile	Own profile	Own profile and user records	Displays account information.

Logout	Yes	Yes	Ends the authenticated session.

Table 6: Role-Based Navigation

5.3 E-KAP Login Interface

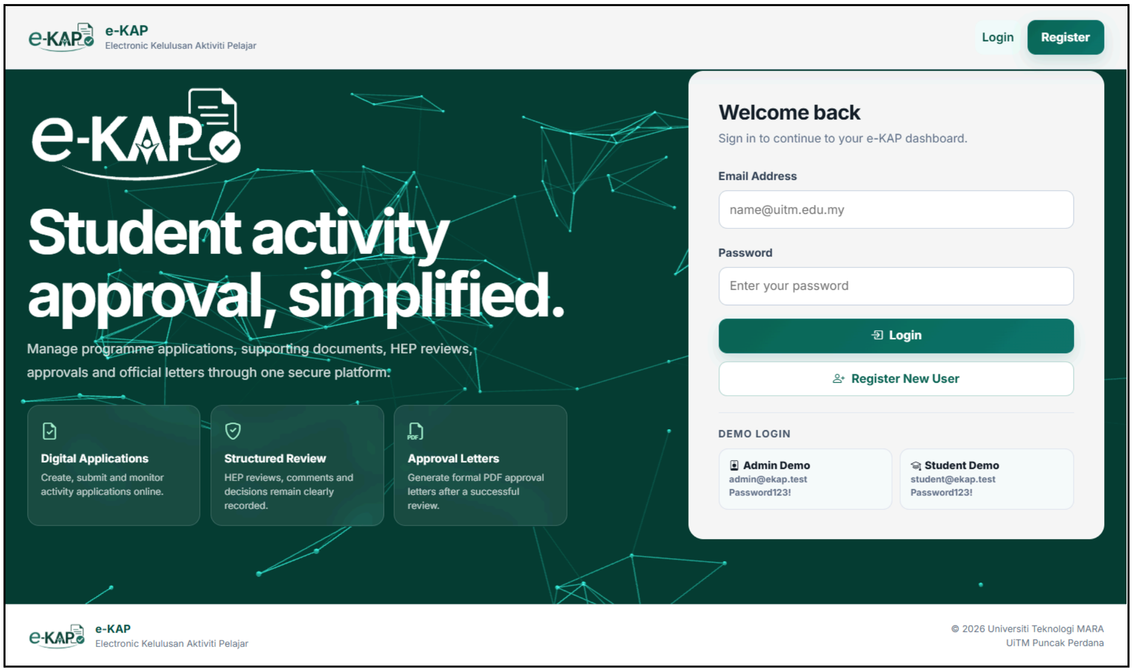

Figure 3: E-KAP Desktop Login Interface

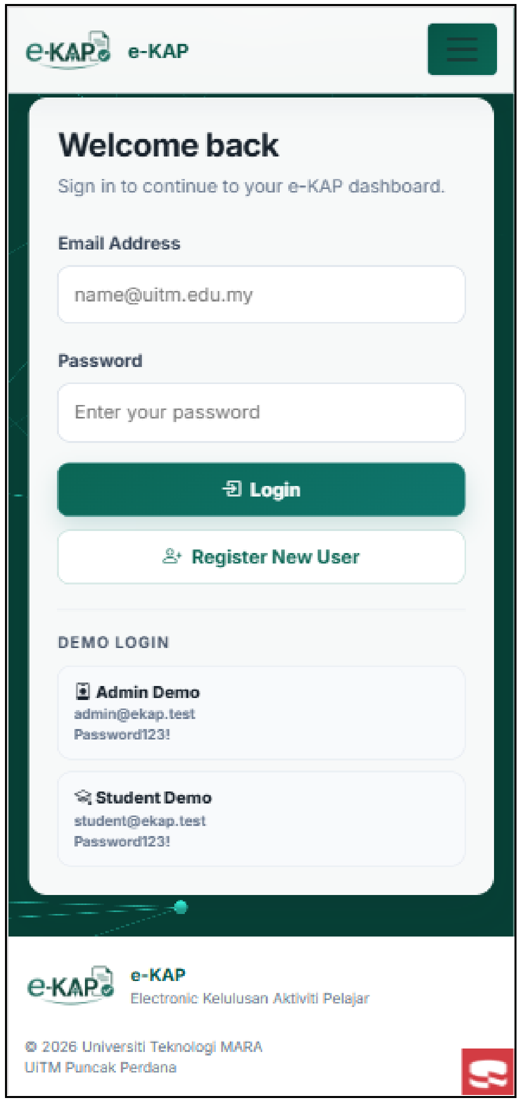

Figure 4: E-KAP Login Interface for mobile view

5.4 Student Dashboard

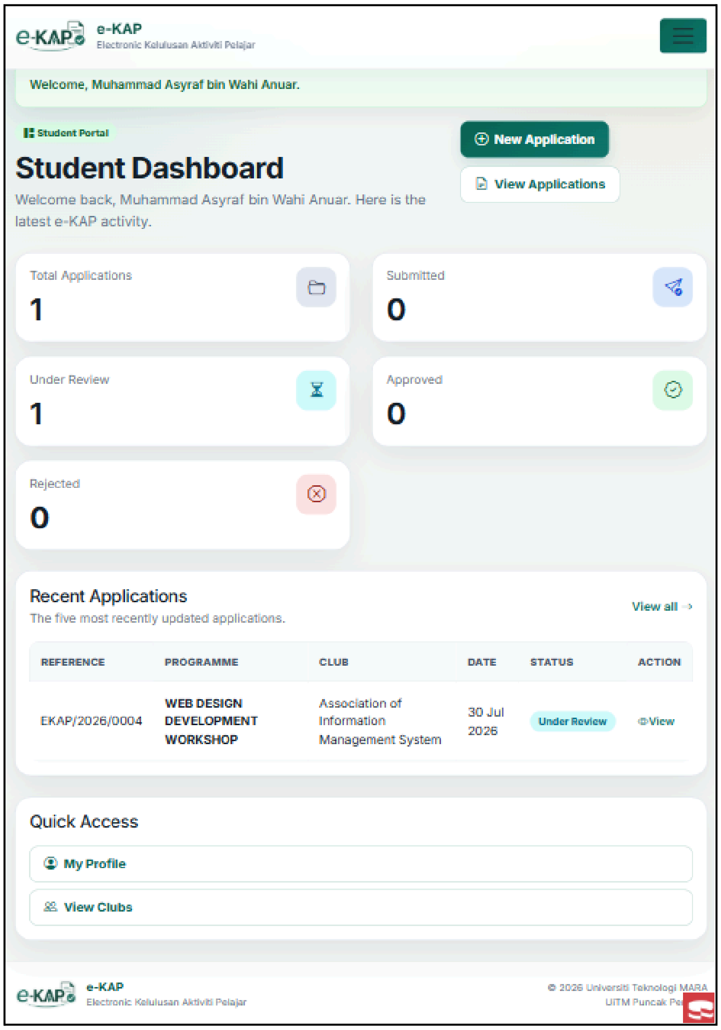

Figure 5: Student Dashboard Mobile View (iPad Air)

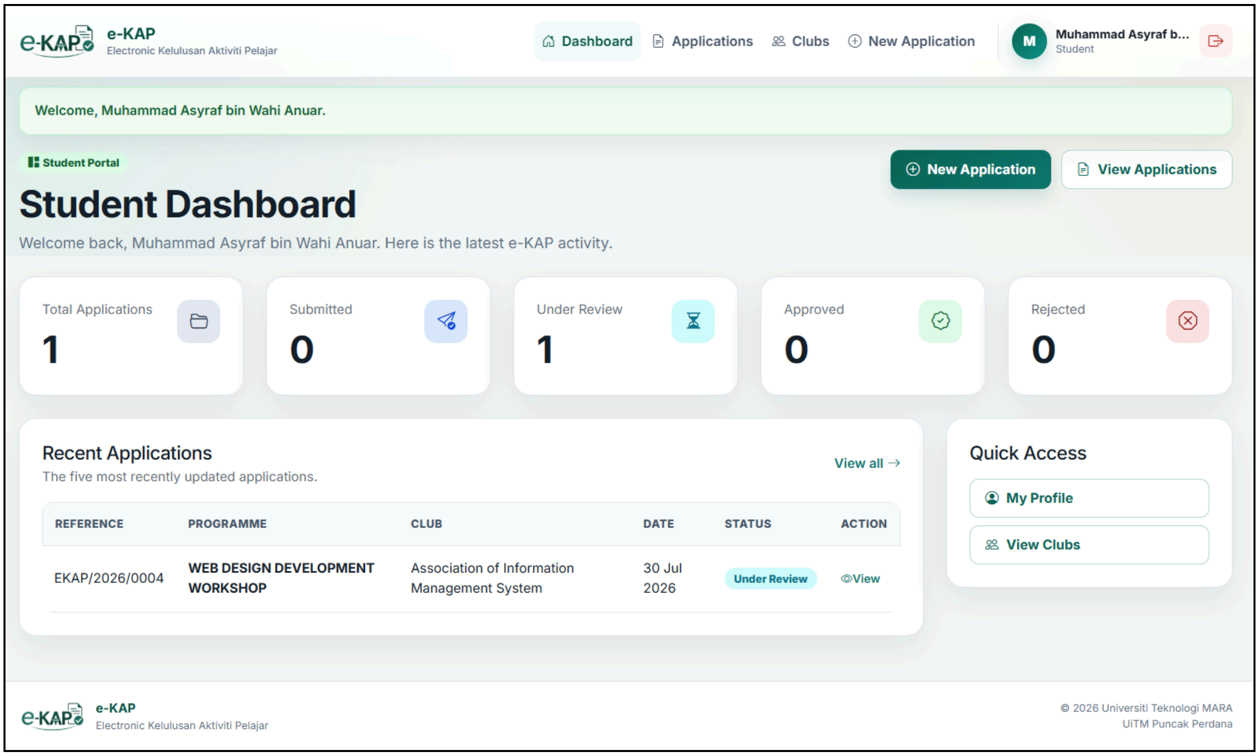

Figure 6: Student Dashboard Desktop View

5.5 HEP Administration Dashboard

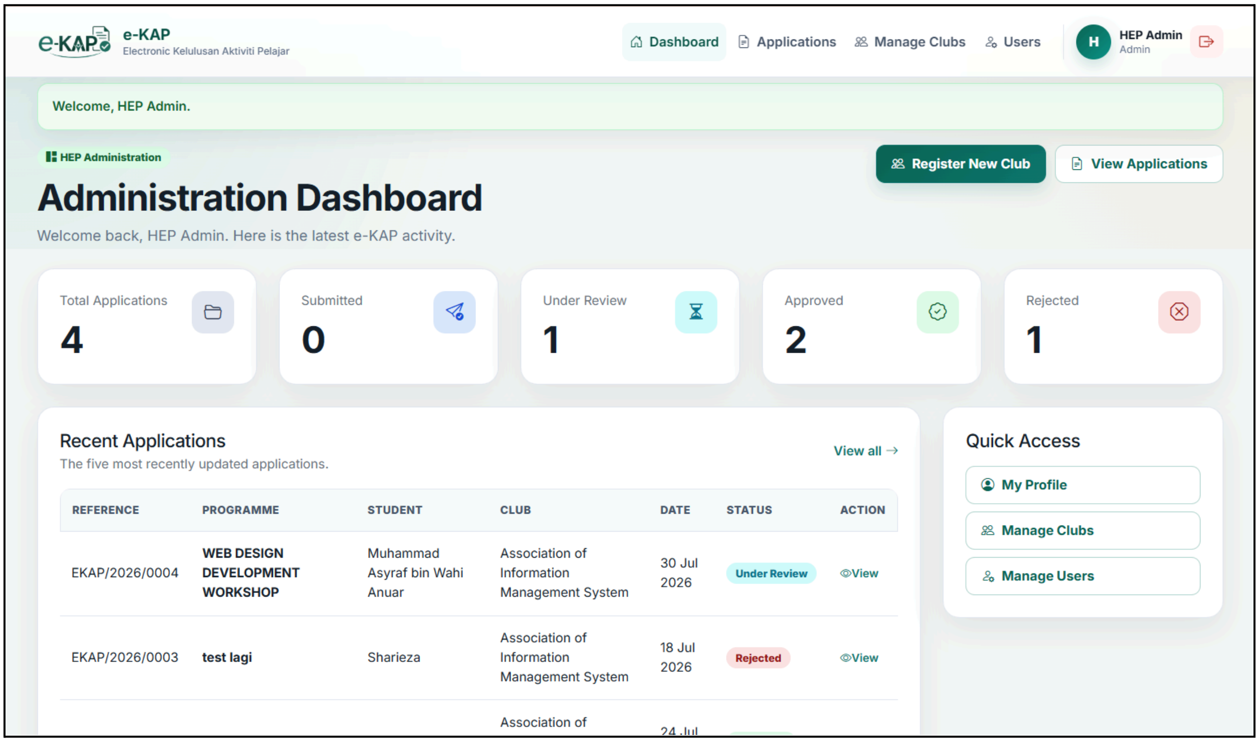

Figure 7: HEP Administration Dashboard Desktop View

6.0 FEATURES & FUNCTIONALITIES

6.1 User Registration and Authentication

New users may register as a student or administrator. The registration form collects the user's full name, email address, password, role, faculty, role-specific identifier, and phone number. The interface dynamically displays a Student ID field for student accounts and a Staff ID field for administrator accounts. Passwords must contain at least eight characters and are stored using CakePHP's DefaultPasswordHasher rather than plain text.

Student registrations are activated immediately. A publicly registered administrator account is stored as inactive and must be activated by an existing administrator before it can be used. During login, the system accepts only active accounts, verifies the hashed password, renews the session, and stores the authenticated user's ID, name, email, and role in the session.

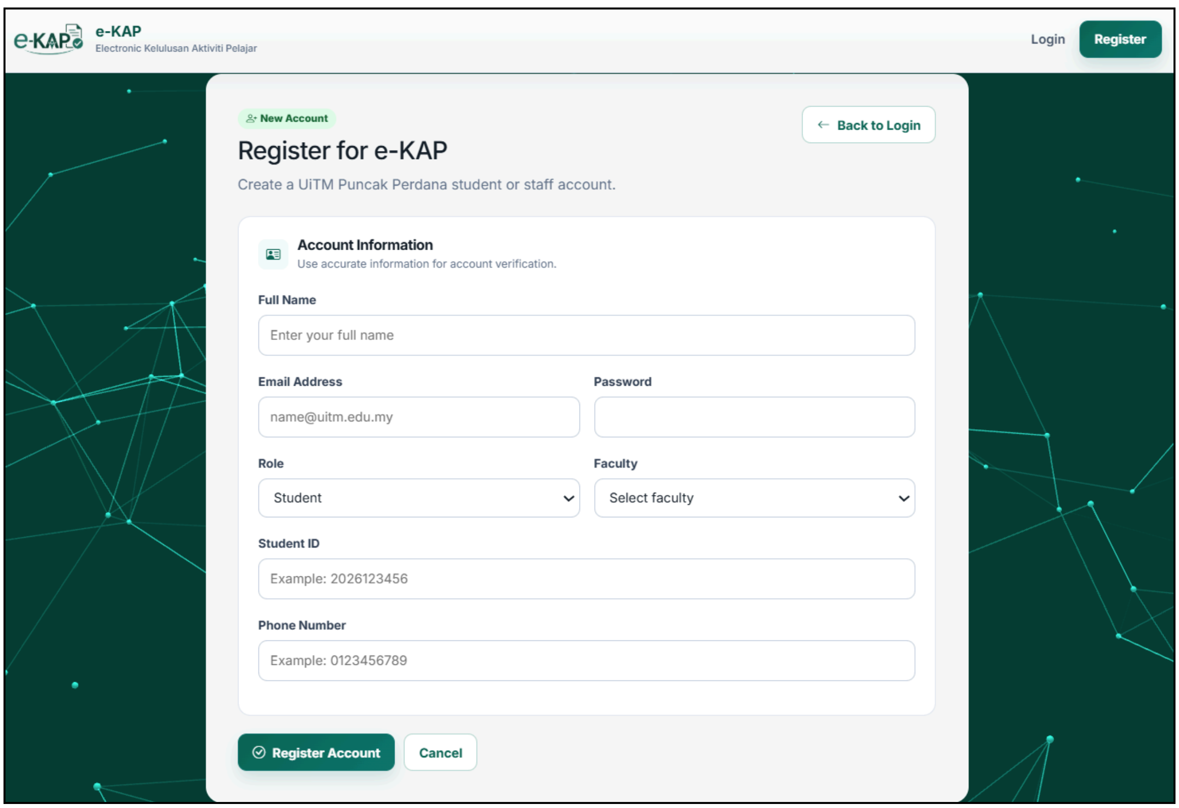

Figure8: Registration form showing role selection and Student ID/Staff ID fields

6.2 Role-Based Dashboard

The dashboard presents five key application statistics, which are total applications, submitted applications, applications under review, approved applications and rejected applications. A student sees counts and recent records only for applications created by that student. An administrator sees statistics across all applications. The dashboard also lists the five most recent applications and provides direct access to common actions.

6.3 Club and Society Management

Administrators can register clubs and societies by entering the club name, unique registration number, faculty, advisor name, advisor email and status. Active clubs automatically appear in the student application form. The system also creates a protected active option called “Others” when required. This option is linked to the database so that the selected club value continues to satisfy the application foreign-key requirement, and administrators are not allowed to delete it.

Students may view registered clubs, while only administrators may create, edit or delete club records. A club that is already connected to application records may not be deleted because referential integrity must be preserved

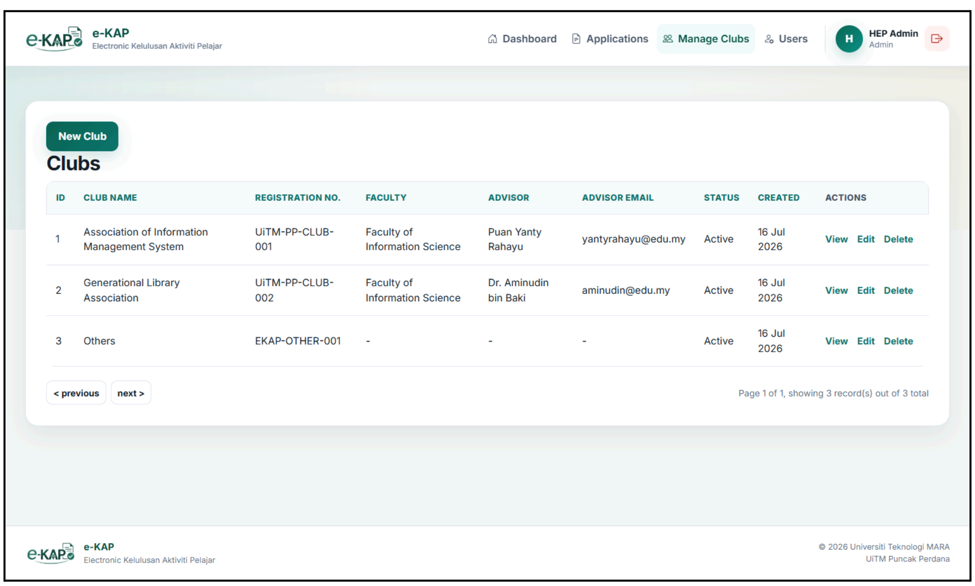

Figure 9: Manage Club Page

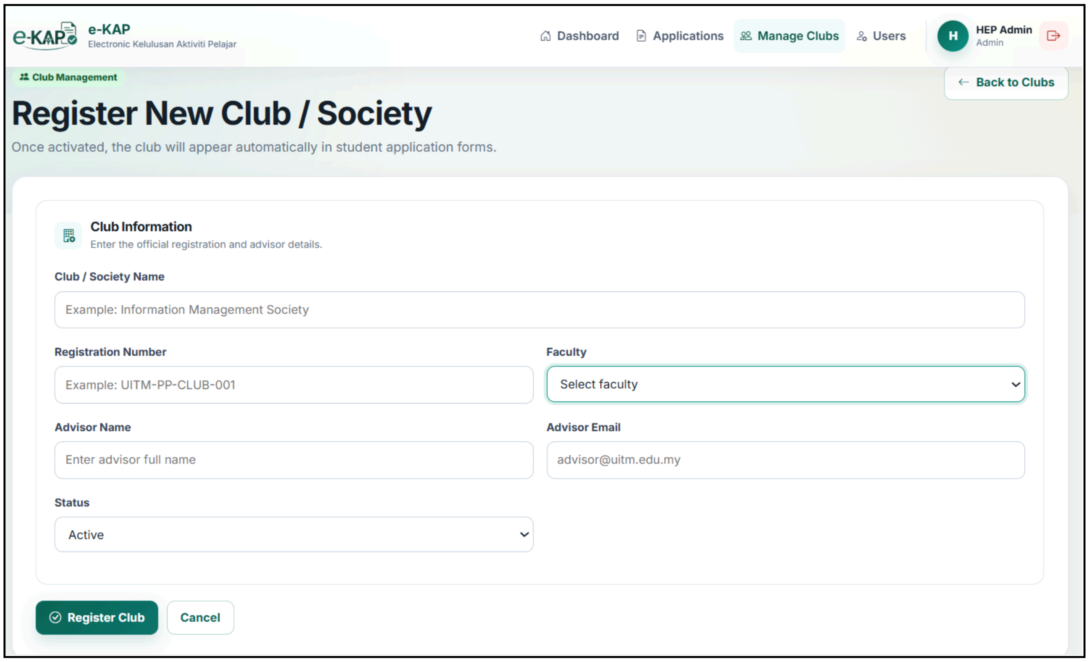

Figure 10: New Club Registration Form

6.4 Programme Application Form

Only student accounts may create programme applications. The form is organised into clear sections and collects the selected club, organiser type, programme title, programme level, programme category, venue, target group, start and end date and time, male and female participant numbers, programme description, objectives, expected outcomes, estimated budget, requested allocation, funding source, risk information and person-in-charge details.

The system automatically calculates the total number of participants by adding the male and female participant values. When a new form is saved, it is stored as a draft. The application user ID is taken from the authenticated session rather than from user-submitted form data. The status, reference number and submission date are also controlled by the server to prevent users from changing protected workflow fields.

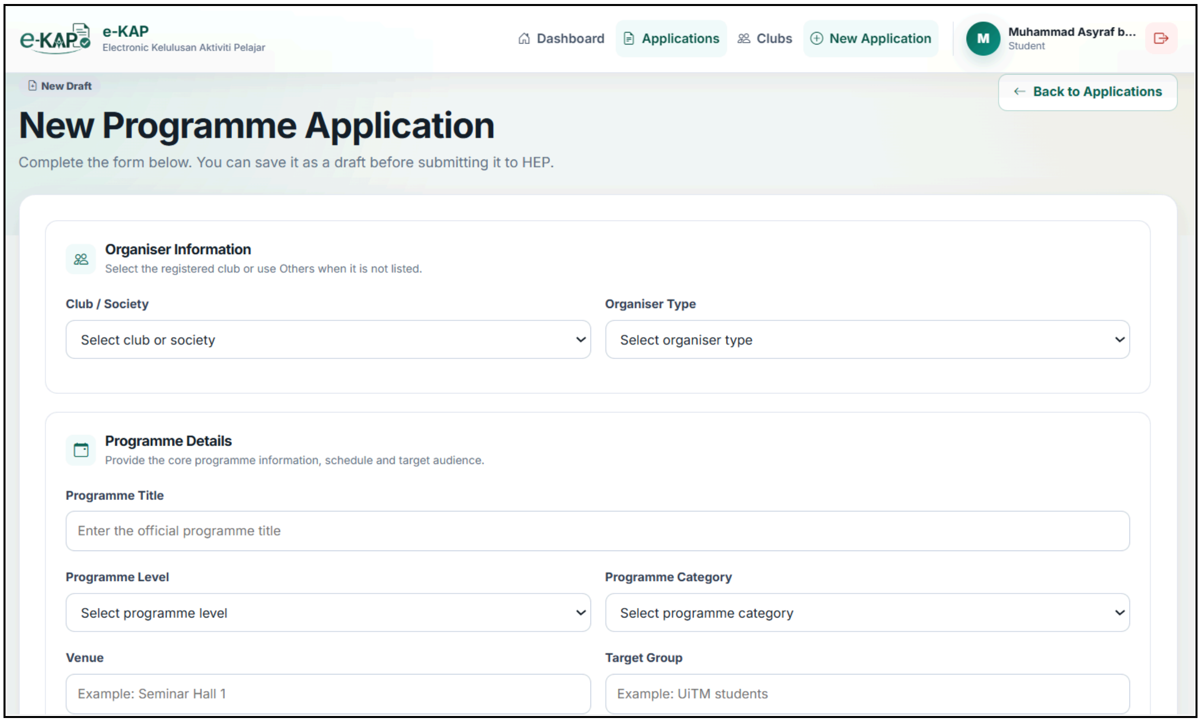

Figure 11: Programme Application Form

6.5 Application List, Search and Filtering

The Applications page displays programme applications in descending order of creation date. Students see only their own records, while administrators see every record. Users may search by programme title, application reference number or club name. A status filter can also be used to display records such as draft, submitted, under review, changes requested, approved or rejected. Each result provides access to a detailed view of the application.

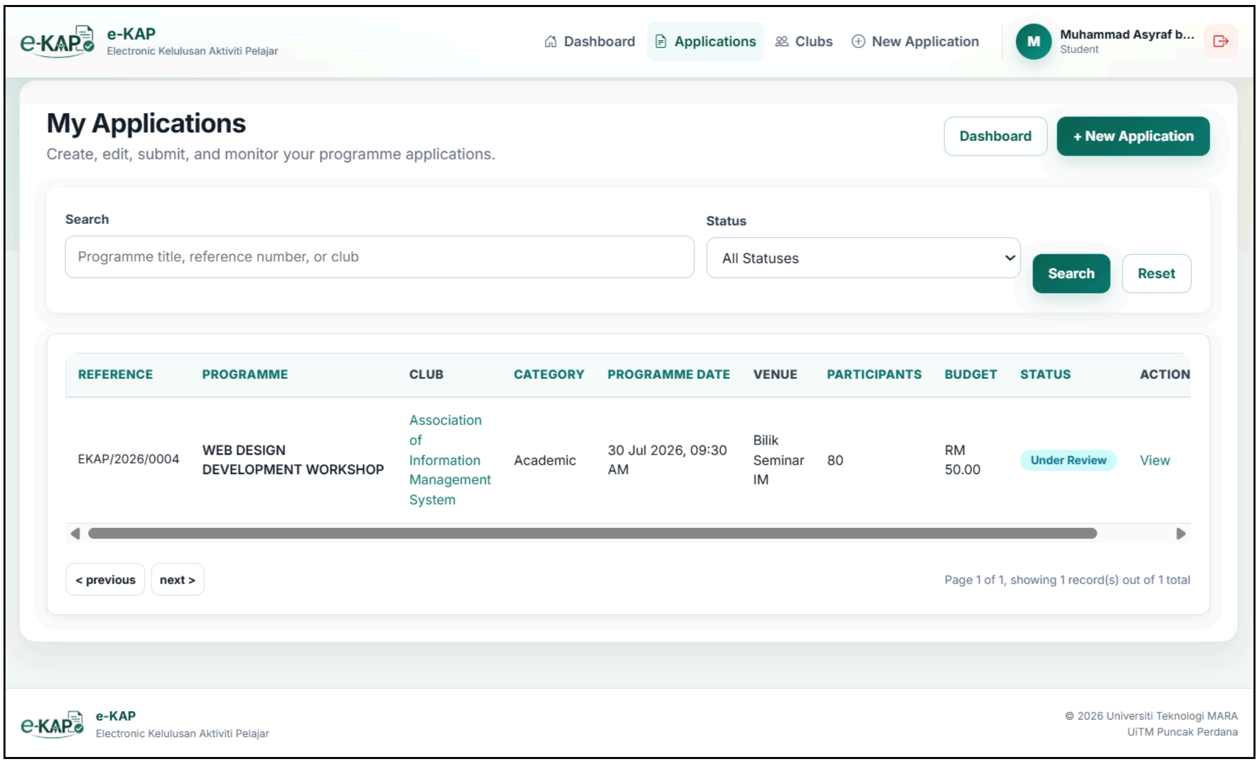

Figure12: Application Search and Status Filtering

6.6 Application Editing, Submission and Tracking

A student may edit an application only when its status is draft or changes requested. The system prevents administrators from changing student programme information because administrative decisions are handled through a separate review function. A student may delete only a draft application. These restrictions help preserve the integrity of applications that have already entered the HEP review process.

When the student submits an eligible application, the system generates a unique reference number such as EKAP/2026/0001, changes the status to submitted and records the submission date and time. The student can then monitor the current status and read the latest HEP comments through the application detail page.

6.7 HEP Review and Decision Management

HEP administrators may review applications with a submitted or under review status. The available decisions are under review, changes requested, approved and rejected. Every decision creates a review record containing the application ID, administrator ID, decision, comments and internal notes. The application status and administrator comment are updated within the same database transaction, reducing the risk of an incomplete update.

Comments are compulsory when an administrator requests changes or rejects an application. When an application is approved, the administrator must enter the approved allocation amount, and the system verifies that the value is not negative and does not exceed the amount requested by the student. The review history remains linked to the application for future reference.

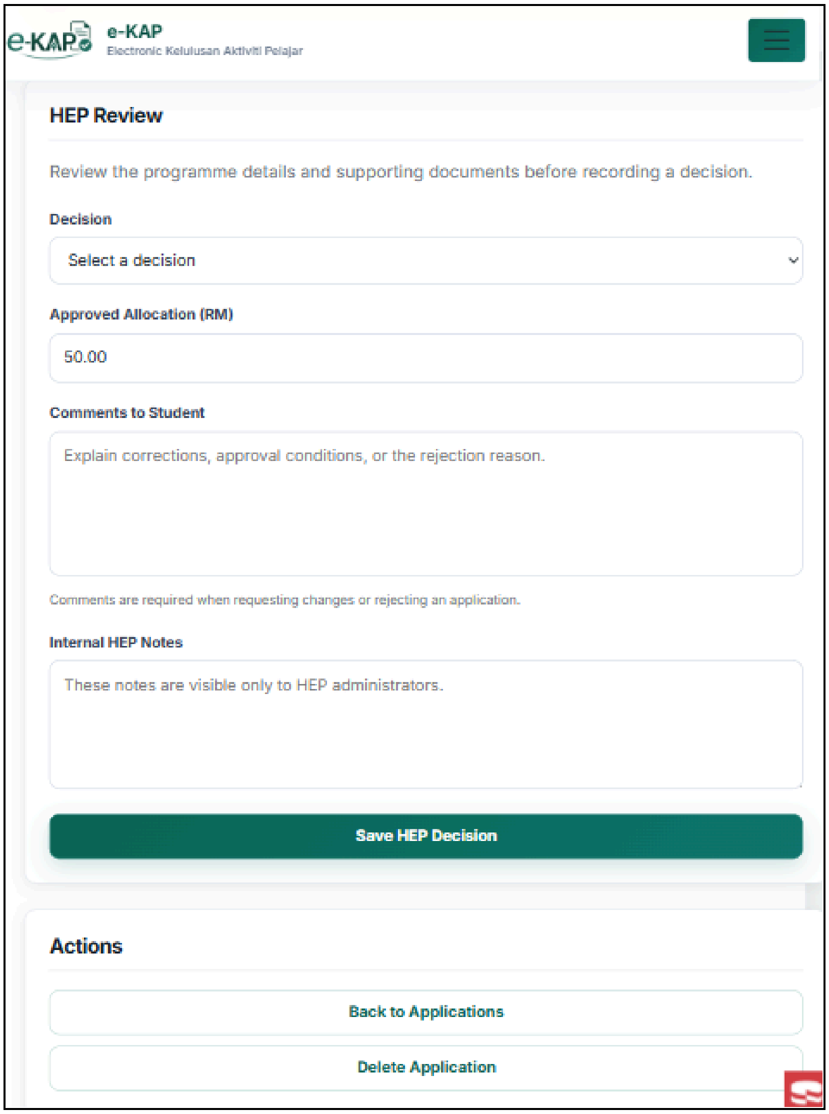

Figure 13: HEP Application Review Interface

6.8 PDF Approval Letter Generation

An official approval letter can be generated only after an application has been approved. The system uses Dompdf to render an A4 portrait PDF containing the application reference number, programme information, budget information, approval date and institutional details. The PDF also includes UiTM and e-KAP branding and page numbering. Students may download only the approval letter for their own application, while administrators may access approved application letters as part of their management role.

UNIVERSITI TEKNOLOGI MARA

e-KAP

Universiti Teknologi MARA

Fakulti Sains Maklumat

Bahagian Hal Ehwal Pelajar
Kampus Puncak Perdana

Rujukan : EKAP/2026/0002
Tarikh : 16 Julai 2026

Sharieza
No. Pelajar: 2025234567
Wakil Association of Information Management System
Faculty of Information Science
Universiti Teknologi MARA
Kampus Puncak Perdana

Assalamualaikum dan Salam Sejahtera,

Saudara/Saudari,

KEPUTUSAN PERMOHONAN KELULUSAN DAN PERUNTUKAN KEWANGAN BAGI PENGANJURAN PROGRAM

TEST PROGRAM

Dengan hormatnya perkara di atas adalah dirujuk.

2. Sukacita dimaklumkan bahawa permohonan penganjuran program **Test Program** yang dikemukakan melalui Sistem e-KAP telah diteliti dan **DILULUSKAN** oleh pihak Hal Ehwal Pelajar.

3. Program tersebut dijadualkan berlangsung pada **24 Julai 2026, 08:50 AM** hingga **24 Julai 2026, 01:50 PM** bertempat di **Dewan Seminar IM**. Peruntukan yang diluluskan adalah sebanyak **RM50.00**. Ringkasan kelulusan adalah seperti di Lampiran A.

4. Kelulusan ini tertakluk kepada syarat-syarat berikut:

i. Penganjuran hendaklah mematuhi peraturan universiti, undang-undang yang berkuat kuasa, serta nilai dan tatakelakuan pelajar.

ii. Sebarang pindaan terhadap tarikh, tempat, skop, tentatif, perbelanjaan atau kertas kerja asal hendaklah mendapatkan kelulusan bertulis daripada pihak Hal Ehwal Pelajar terlebih dahulu.

iii. Sebarang tajaan dalam bentuk wang atau barangan hendaklah diuruskan melalui saluran kewangan universiti yang telah diluluskan.

iv. Penasihat, penganjur dan pegawai pengiring bertanggungjawab memastikan pengurusan program, kebajikan peserta serta keselamatan pelajar dilaksanakan dengan sewajarnya.

v. Program di luar kampus atau program berisiko hendaklah memperoleh semua kebenaran tambahan, perlindungan dan dokumen keselamatan yang diperlukan sebelum pelaksanaan.

Fakulti Sains Maklumat, Universiti Teknologi MARA
Kampus Puncak Perdana, 40150 Shah Alam, Selangor

e-KAP - Kelulusan Aktiviti Pelajar
Dokumen dijana secara elektronik
Halaman 1 / 4

Figure 14: PDF document

6.9 User Account Management

Administrators can list, create, view, edit and delete user accounts. They may also change an administrator registration from inactive to active. Non-administrator users may view and edit only their own profile and are not permitted to change protected account fields such as role, account status or staff number. The system also prevents an administrator from deleting the administrator account currently being used.

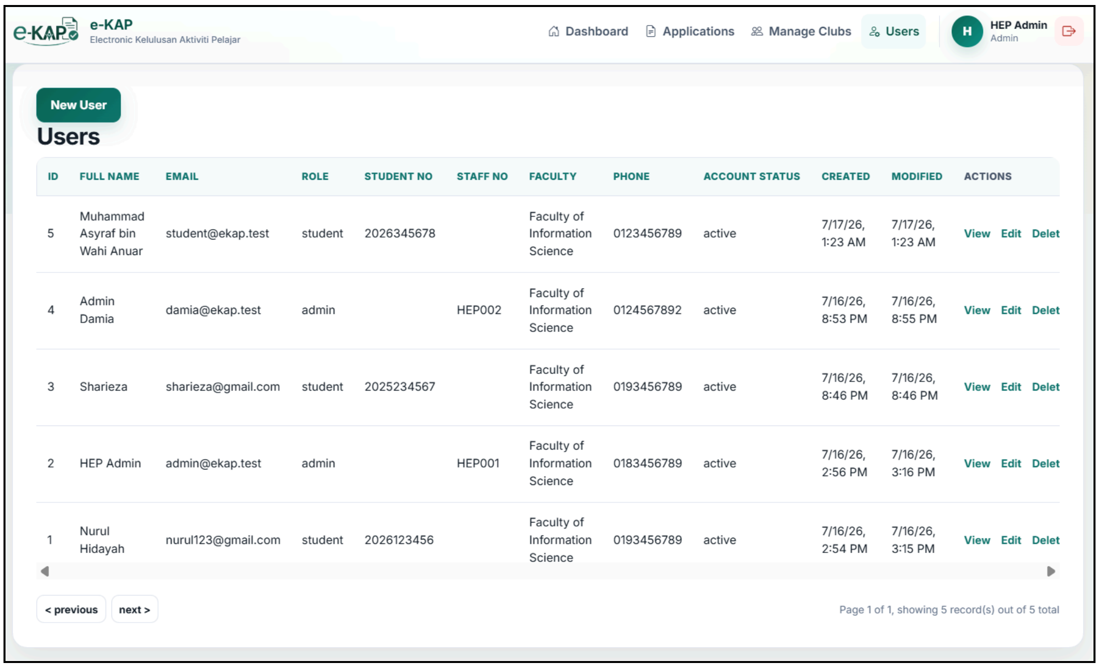

Figure 15: Administrator User Account Management

6.10 Data Validation and Security Controls

- All standard pages require an authenticated e-KAP session; only login and registration are public.
- Passwords are hashed before being stored in the database.
- Login is restricted to active accounts.
- The session is renewed after a successful login.
- Student access to application records is checked by user ID to prevent URL manipulation.
- Role checks prevent students from using administrator functions and administrators from editing student programme details.
- Server-controlled fields such as user_id, status, reference_no and submitted_at cannot be overwritten through the application form.

- Database validation enforces required fields, valid option lists, unique identifiers and valid foreign keys.
- Approval-letter access is restricted to approved applications and authorised users.

7.0 WORKFLOW OF FORM AND CRUD CYCLE

7.1 Programme Application Workflow

The programme application process begins when a student registers or logs in and opens the New Application page. The student completes the form and saves it as a draft. While the application is still a draft, the student may read, update or delete it. After submission, the application receives a reference number and becomes available to HEP administrators for review. An administrator may mark the record as under review, return it for changes, approve it or reject it. A changes-requested application returns to the student for correction and resubmission. An approved application allows the official PDF approval letter to be generated.

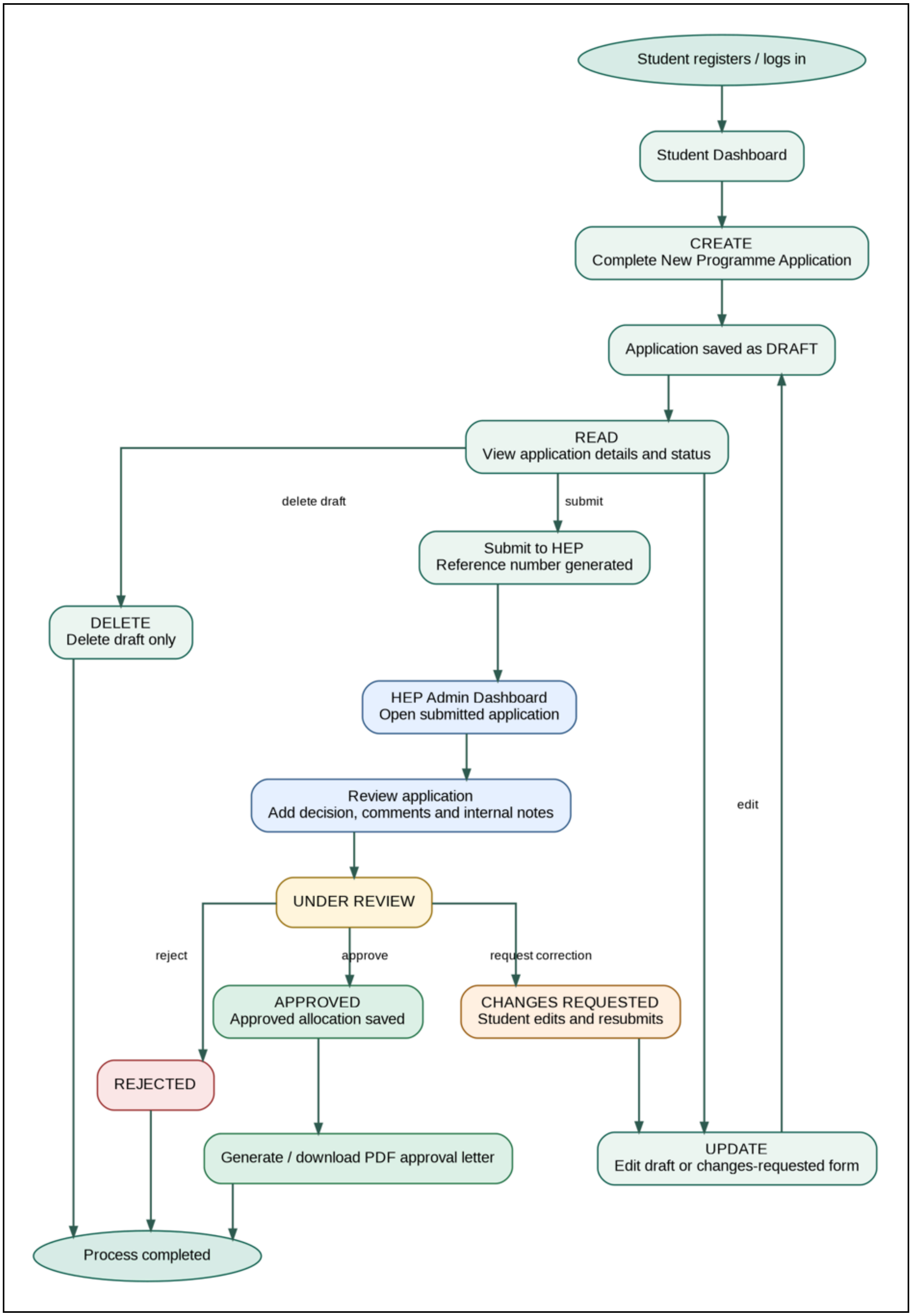

Figure 15: E-KAP System Application Workflow

7.2 CRUD Cycle Explanation

CRUD Operation	Implementation in e-KAP
Create	Student creates an application draft. Administrators create club and user records. Review and document records can also be created and linked to applications.
Read	Users view dashboards, application lists, detailed application information, club records, user profiles, document records and review history according to their role.
Update	Students update draft or changes-requested applications. Administrators update club and user records and update application status through the review process.
Delete	Students delete only their own draft applications. Administrators may delete eligible club and user records. The protected “Others” club and the current administrator’s own account cannot be deleted.

Table 7: CRUD Application In E-KAP

8.0 TEAM ROLES & CONTRIBUTION

Team Member	Position	Responsibilities
Wan Norsyafiqah	Project Manager	Planning, Coordinating tasks, preparing documentation and reports.
Nurhidayah	Front-End Developer	Designing user interface using HTML and CSS.
Nurul Sharieza	Back-End Developer	Developing system functionalities using CakePHP, PHP, and MySQL.
Nurul Aina' Damia	System Analyst	Gathering requirements, designing system flow and database structure.

Table 8: Team Roles and Responsibilities

9.0 CONTACT INFORMATION (SUPPORT)

Item	Information
System / Project	e-KAP: Electronic Kelulusan Aktiviti Pelajar
Support Contact Person	Nurhidayah Muhammad Adzmin @ Ajiun
Email Address	2024696368@student.uitm.edu.my

10.0 CONCLUSION & REFLECTION

The development of e-KAP (Kelulusan Aktiviti Pelajar) has the primary objective of providing a website system to improve the student activity approval process. The system was developed to replace the traditional manual procedure, where students were required to print application forms, complete them manually and submit them to the relevant office for approval. By transforming this process into a digital platform, e-KAP offers a more efficient, organized and user friendly solution for both students and staff.

Through the system, students can conveniently log in, submit activity approval applications, upload supporting documents, and monitor the status of their applications online. At the same time, staff members are provided with an administrative interface that enables them to review submitted applications, verify supporting documents, improve record management and shorten the overall approval process. The centralized database also ensures that application information is stored securely and can be accessed whenever necessary.

Overall, e-KAP demonstrates how information technology can be used to improve administrative processes within educational institutions. The implementation of this website system not only increases efficiency and transparency but also enhances communication between students and staff. The project has successfully met its objectives and provides a practical solution that can contribute to better management of student activity approvals.

Reflection

The development of e-KAP has been a valuable learning experience throughout this project. It provided the opportunity for this project. It provided the opportunity to apply theoretical knowledge gained during the course of development of real world website system. Throughout the project, we used technologies such as PHP, MySQL, Bootstrap, Laragon and Visual Studio Code, were used to design, develop and implement the system. This experience both practical knowledge and practical skills in web development, database management and user interface design.

This project also highlighted the importance of careful planning, proper documentation, and systematic testing throughout the software development lifecycle. Preparing the Entity Relationship Diagram (ERD), designing the user interface, creating workflow processes, and performing testing all contributed to the successful completion of the project. Each stage required attention to detail and continuous improvements to ensure that the final system met the intended requirements.

In conclusion, the development of e-KAP has significantly enhanced knowledge in web system development while improving critical thinking, technical competency, and problem-solving abilities. This project has provided practical experience that will be beneficial for future academic work and professional careers in the field of information systems. It also demonstrates how digital technologies can be effectively applied to solve administrative challenges and improve operational efficiency within educational institutions.